

Manankumar Thakkar

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SUMMARY

Software Engineer experienced in developing scalable, high-performance backend systems with Java, Spring Boot, and RESTful microservices. Skilled in cloud-native data pipelines on GCP and Azure, CI/CD, observability, and building LLM, agentic, and MCP-based tools. Proven in production: cut API latency **60%** and prevented a **6x data collision** during multi-tenant expansion.

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript, TypeScript
Backend: Spring Boot, Microservices, REST APIs, Flask, Kafka, Liquibase, JUnit, Node.js, React
AI / LLM: LLM integration, Agentic systems, MCP / tool use, LangChain, Retrieval-grounded QA, Prompt engineering
Cloud & Data: GCP (BigQuery, GCS), Azure SQL, AWS, Kubernetes, Docker, Redis, Terraform, Tableau
DevOps & Quality: CI/CD, Grafana, Prometheus, Splunk, OpenObserve, SonarQube, Snyk

EXPERIENCE

Walmart Global Tech Bentonville, AR
Software Engineer III 02/2025 – Present

- Architected the Vision Center feature in Java/Spring Boot so the Inventory Reservation Service could set loop policy per location, and led the migration from percentage- to seconds-based loop metrics, expanding advertiser booking flexibility **40%**.
- Diagnosed a production N+1 query bottleneck in the in-store advertising microservices and fixed it with concurrent caching, cutting API latency **60%** and eliminating **10,000+ database calls** per day.
- Caught a key-grain design flaw while integrating multi-tenant international expansion across 7+ partner orgs, and drove a durable country-scoped key over a regional patch, preventing a **6x cardinality collision**.
- Replaced a hand-assembled store-readiness report with an automated Python pipeline (Azure SQL, Redis, BigQuery, Tableau) applying 13 business rules across **4,000+ stores** twice daily; validated to an exact match and fixed **3 latent defects**.
- Built an LLM-powered operations assistant (Python, Flask, LangChain) on Walmart's internal LLM Gateway so engineers could query operational context in chat, with session-scoped memory and response caching.

Software Engineer 07/2024 – 01/2025

- Built Java/Spring Boot features for Store Department Hours and Sam's Club Plus operating hours, extending the store data model to Deli, Bakery, and Service Deli schedules across **1,000+ locations**.
- Implemented location-to-screen-group mapping for new retail formats (Grab-N-Go, Sam's Club Cafe) and filtering across stores, inventories, and device-commands REST APIs for more precisely targeted ad placements.

George Mason University Fairfax, VA
AI Research Assistant 07/2023 – 07/2024

- Applied LLM tools (ChatGPT) to build research tooling that automated and streamlined research workflows.
- Provisioned and managed research infrastructure as code with Terraform, supporting reliable research environments.

Nokia Remote, US
Test Automation Engineer (Co-op) 09/2022 – 06/2023

- Automated testing with Python suites for high-throughput systems, reducing manual testing effort **50%**.
- Integrated tests into Jenkins CI/CD pipelines, cutting defect density **20%** and improving code quality metrics **35%**.

Let It Wag Mumbai, India
Full Stack Developer 06/2019 – 05/2020

- Engineered Node.js services and REST APIs handling real-time data for **13K+ users**, improving backend efficiency **20%**.
- Developed a responsive React UI supporting 100+ user journeys and delivered 90% of projects on time.

AI PROJECTS

AuthForge | Python, MCP, Spring Boot, Istio

- Designed an MCP server that remediates missing-authentication findings in Walmart Java services via config alone.
- Implemented 9 tools spanning diagnosis, endpoint scanning, consumer discovery, config generation, and validation.

Wayfinder (Walmart Global Tech Hackathon) | Python, LLM orchestration, agent routing

- Co-built an agentic system that triages Slack, email, and calendar by intent and routes work to sub-agents.
- Owned the routing layer, query sanitization, and extensibility framework, with one-click human approval gates.

EDUCATION

George Mason University, Fairfax, VA – *Master of Science in Computer Science* 08/2021 – 05/2023
Coursework: Data Structures and Algorithms, Software Modeling and Architectural Design, Data Mining, Advanced NLP